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Breaking the glass ceiling for a sustainable future: the power of women on corporate boards in reducing ESG controversies

Reducing ESG controversies

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Abstract

Purpose – This study aims to investigate the relationship between board gender diversity and environmental, social and governance (ESG) controversies and to determine if a critical mass of female directors has a significant impact on ESG performance.

Design/methodology/approach – The study analyzes a sample of non-financial companies from 13 European countries between 2004 and 2021. The primary method used to reach conclusions was the pooled ordinary least squares regression. Additionally, the study used supplementary techniques such as alternative measurement, sub-sample analysis and two-stage least squares to enhance its reliability.

Findings – The results indicate that a higher representation of women on boards is correlated with a reduction in the number of ESG controversies, particularly when there are three or more female directors. Furthermore, the relationship between board gender diversity and ESG controversies may be affected by factors such as industry, governance and a company's environmental performance.

Practical implications – This study suggests that increasing women's representation on boards may mitigate ESG controversies and improve firm reputation and performance, especially in industries with high ESG risks. Policymakers can support this through policies, targets, training and inclusive practices. The findings also inform investors and stakeholders of the relationship between board gender diversity and ESG controversies.

Originality/value – This study expands the understanding of the relationship between board gender diversity and sustainable accounting and finance. It focuses on the effect that having female board members has on corporate policies, which is significant for shaping global policies that promote diversity on boards.

Keywords Corporate governance, Board gender diversity, ESG controversies, Environmental impact

Paper type Research paper

1. Introduction

The consequences of climate change are already being felt globally, and unless action is taken soon, the future effects will be much more severe (Issa and Zaid, 2023). Europe, like other regions in the world, is also being affected by global warming, as it is the third most populous continent on Earth. All countries in Europe are experiencing the effects of global warming in some form, including air pollution, droughts, wildfires and rising sea levels. In response to the challenges of climate change, many countries have announced targets to reduce their greenhouse gas (GHG) emissions and some have even made “net-zero commitments,” which mean they aim to reduce GHG emissions as much as possible and

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balance between the amount of GHG produced and the amount removed from the atmosphere. For instance, the European Union and its 27 member states have committed to reducing their GHG emissions to net zero by 2050, with the goal of achieving a minimum 55% reduction in emissions compared to 1990 levels by 2030.

However, it is important to note that the majority of GHG emissions are not produced by individuals, but rather by companies and industries. These emissions are often associated with energy production, transportation and industrial processes. According to a report by the Carbon Disclosure Project (CDP) [1] in 2017, just 100 publicly listed companies were found to be responsible for more than 70% of all climate-related emissions globally. This highlights the fact that a small number of large companies are responsible for a significant proportion of GHG emissions, and thus, their actions will play an important role in achieving global emission reduction targets.

Nevertheless, many companies still do not disclose their GHG emissions. In response, many countries have begun implementing mandatory disclosure rules for corporate climate-related information, particularly for GHG emissions. For instance, in 2019, the UK government established regulations mandating that publicly traded companies must disclose information on risks related to climate change and their GHG emissions. In addition, the USA government has implemented similar regulations that will require companies to disclose information about their GHG emissions and climate-related risks starting April 2022. These mandatory disclosures can play a crucial role in helping drive the transition toward a green economy by making companies more accountable for their emissions and encouraging them to reduce their environmental impact.

It is widely acknowledged by scholars, practitioners and policymakers that mandatory reporting of GHG emissions alone is not sufficient to address the climate crisis and other factors also play a role in a company's environmental, social and governance (ESG) performance. Research has shown that the composition of a company's board of directors can have an impact on corporate decisions related to ESG issues and can help to reduce controversies related to these matters (Issa and Fang, 2019). For instance, board gender diversity is found to be linked to improved waste management and recycling (Gull *et al.*, 2023), reduced biodiversity impact (Haque and Jones, 2020), decreased environmental emissions (Kyaw *et al.*, 2022) and enhanced eco-innovation (Issa and Bensalem, 2022). Board gender diversity is also associated with improved accounting misreporting (Wahid, 2019), earnings quality (Srinidhi *et al.*, 2011) and accounting misreporting (Cumming *et al.*, 2015). However, there has not yet been a study that specifically examines the relationship between board gender diversity and reduced controversies related to ESG issues. ESG controversies can have significant negative impacts on society, as well as on a company's financial performance (Sheehan *et al.*, 2023). These controversies can lead to devastating losses in shareholder value (Aouadi and Marsat, 2018). This is why, it is important for companies to address and mitigate these types of issues to protect their reputation and ensure long-term financial stability. Additionally, as more and more investors are becoming aware of the importance of ESG factors and are incorporating these factors into their investment decisions, companies that have a good track record on ESG issues are more likely to attract investment and be perceived as more valuable by the market (Sheehan *et al.*, 2023).

Given the growing significance of environmental and social matters (Issa *et al.*, 2021), it is important and relevant to examine the connection between gender diversity on boards and the controversies that companies may face regarding their ESG performance. This research provides valuable understanding of how diversity among leaders can impact a company's reputation and its impact on society, and as ESG becomes increasingly relevant to investors and consumers, understanding the relationship between diversity on boards and a company's performance in these areas is crucial for both the private and public sectors.

In this study, we argue that companies with more women on their boards of directors are less likely to experience controversies related to ESG issues. This hypothesis is based on the gender socialization theory, which suggests that men and women may have different approaches to decision-making and risk-taking, and the diversity theory, which argues that diversity in decision-making groups can lead to better outcomes. Additionally, the literature on overconfidence suggests that men may be more prone to taking excessive risks, which could lead to ESG controversies. Therefore, it is hypothesized that a more gender-diverse board may be less likely to experience ESG controversies.

To empirically test our study's framework, we rely on a panel data set of non-financial enterprises operating in 13 European nations between 2004 and 2021. The empirical findings show strong support for our theoretical assumption that board gender diversity is negatively and significantly related to ESG controversies. Additionally, the findings suggest that to have a significant impact on ESG performance, a board should have a minimum of three women, which is in line with the critical mass theory.

We use the ordinary least squares (OLS) technique as a baseline model to examine the link between board gender diversity, using the proportion of female directors on the board, and the ESG controversies score. However, it could be that firms that have implemented strict ESG measures are the ones that have a greater proportion of women on their boards. This implies that firms that have already invested in environmental performance may also be more inclined to prioritize diversity in decision-making, including the appointment of more female directors. This could mean that our results may be affected by the issue of reverse causality or simultaneity. We address the issue of endogeneity using some techniques. First, we use a one-year lagged board gender diversity variable to determine if our findings remain consistent over time. Second, we use the two-stage least squares (2SLS) method to correct endogeneity problem. Furthermore, we use a different measure of ESG controversies and subsample tests to discover factors that could confound the results.

The results of this study add to the existing body of literature in multiple ways. First, this study broadens the research on board gender diversity and sustainability accounting and finance by examining the impact of female board directors on corporate policies, which is important for the development of global policies that advocate for board gender diversity. Previous studies have established a connection between board gender diversity and an increase in positive ESG initiatives (Bravo and Reguera-Alvarado, 2018; Kyaw *et al.*, 2022; Liu, 2018; Lu and Herremans, 2019; Marchini *et al.*, 2022). However, there is a gap in the literature as no previous research has examined the relationship between board gender diversity and controversies related to ESG issues. This study uses the ESG controversies score from the Eikon Refinitiv database as the primary measure of a company's ESG performance and presents new evidence showing that board gender diversity is related to a reduction in ESG controversies. Second, this study supports the critical mass theory. The findings indicate that when a board has three or more women, there is a positive impact on the company's ESG controversies score. These results align with previous studies (Joecks *et al.*, 2013; Konrad *et al.*, 2008; Torchia *et al.*, 2011). Overall, the study provides further evidence that supports the critical mass theory, which suggests that having a certain number of women on the board is necessary for them to have a positive impact on the company. Third, this study looks into the possible factors that affect the connection between board gender diversity and ESG controversies. The results suggest that companies with strong governance and high environmental performance may have a stronger relationship between board gender diversity and ESG controversies score. These findings can be used to inform strategies for promoting board gender diversity and reducing ESG controversies.

2. Literature review and hypothesis development

The board of directors plays a crucial role in creating and implementing a firm's ESG strategies and policies that guide a firm's actions and decisions in managing its impact on the environment and community. It has been suggested by several theoretical and empirical perspectives that women may have an advantage over men when it comes to making decisions related to ESG matters, as they are more likely to take a proactive approach to avoid any violations and ensure compliance.

The gender socialization theory suggests that individuals, due to the different way they are raised and the different experiences they have, develop distinct perspectives and strategies when it comes to making decisions. It is believed that men and women may have dissimilar methods of thinking and problem-solving as a result of their upbringing and societal interactions (Gilligan, 1982). Women are socialized to be more attuned to the needs of others and to exhibit greater concern for ethical considerations (Eagly *et al.*, 1990; Eagly and Johannesen-Schmidt, 2001). This may be reflected in their decision-making and leadership style (Rosette and Tost, 2010). There is empirical evidence that suggests there are gender differences in ethical decision-making (Dawson, 1997; Loo, 2003; Lund, 2008; Mason and Mudrack, 1996).

For example, Garcia-Sánchez *et al.* (2015) show that firms with a higher proportion of female board directors may be more likely to have stronger and more comprehensive ethical codes. This is because gender diversity on boards can lead to a more thorough examination of potential ethical issues (Zalata *et al.*, 2018), a greater consideration of the interests of all stakeholders (Lu and Herremans, 2019), fewer environmental violations (Liu, 2018) and lower levels of carbon emissions (Nuber and Velte, 2021).

According to Adams and Funk (2012), female directors tend to be less focused on achieving power and prestige compared to their male counterparts. Instead, they are characterized by having strong traits of benevolence and universalism. These traits embody a deep understanding and protection for the well-being of society and the environment (Galbreath, 2018). Similarly, empirical evidence shows that firms with more women on their boards are more likely to donate to charitable causes (Lin *et al.*, 2018) and to implement environmental and social programs that address issues such as environmental sustainability and community development (Post *et al.*, 2015).

In light of gender socialization and ethicality theories, the role of women in improving ESG is therefore twofold: first, their different values (e.g. benevolence and universalism) and ethical behaviors may lead them to prioritize ESG issues more highly, and second, their presence on boards and in leadership roles can serve as a catalyst for greater attention to and action on ethical issues such as ESG.

According to the diversity theory, the inclusion of different perspectives and backgrounds among those making corporate decisions can improve the outcome of the decision-making process (Siciliano, 1996). Having a balance of both male and female directors on a corporate board can bring unique cognitive traits to the decision-making process as a result of the different expertise and skills that are associated with their gender characteristics (Katmon *et al.*, 2019). According to Ferreira (2010), having a diverse demographic makeup in the boardroom can lead to better decision-making outcomes. This is achieved through several mechanisms such as the presence of a wider range of perspectives, which leads to more innovative problem-solving and the generation of multiple solutions; having access to a wider range of resources and networks, which leads to a better understanding of the marketplace; increased comprehension of the needs and concerns of different stakeholders; and an enhancement of the firm's reputation and credibility.

Furthermore, [Zalata et al. \(2022\)](#) suggest that gender diversity on a board can lead to a reduction in the level of trust among board members. This is because having a diverse group of individuals with different backgrounds, experiences and perspectives can lead to more scrutiny and questioning of each other's actions and decisions ([Cumming et al., 2015](#)). This increased level of scrutiny can lead to improved monitoring of the board's actions and decisions, as individuals are more likely to hold each other accountable and question any potential problems or issues that may arise ([Wahid, 2019](#)). Such benefits of board gender diversity can lead to less fraud and financial misconduct ([Arnaboldi et al., 2021](#); [Cumming et al., 2015](#)).

In light of the above discussion, we argue that having a diverse gender representation on the board can lead to better decision-making when it comes to ESG controversies. This is due to two reasons. First, female directors tend to have different perspectives when it comes to managing environmental risks and concerns, specifically those that affect different groups of stakeholders. This allows the board to consider a wider range of options to minimize environmental impact. Furthermore, having diversity on the board can also potentially allow the board to take into account non-economic considerations, such as the impact of improving environmental performance on the company's reputation, social responsibility and overall sustainability. This can go beyond just considering the impact on the company's financial performance and valuation. Second, having more women on the board reduces the level of trust among board members ([Zalata et al., 2022](#)), which can lead to a more vigilant approach toward ESG policies and practices. As a result, the board would be more likely to take a closer look at potential risks and violations, make more informed decisions to prevent them and minimize environmental risks. This increased level of attention and scrutiny can lead to better decision-making when it comes to ESG matters.

Risk aversion is another angle that can be used to explain the role of female directors in decision-making, particularly in the context of ESG ([Issa, 2023](#)). Research has suggested that women may be more risk-averse than men in certain contexts ([Boulouta, 2013](#)), which could lead them to be more cautious and vigilant when it comes to potential environmental and social risks associated with corporate decision-making ([McGuinness et al., 2017](#)). This increased level of attention to risk can lead to more thorough and thoughtful decision-making and can help to minimize the potential negative impacts of business activities on society and the environment. Therefore, we argue that the presence of female directors on boards can be seen as an important aspect when it comes to handling ESG controversies because of their ability to anticipate and mitigate potential risks, and their ability to navigate uncertain and complex situations:

H1. Having women on the board decreases a firm's ESG controversies.

While research suggests that women can have a positive impact on ESG strategies and practices, having just one female director on the board may not have a significant impact on corporate decision-making ([Kanter, 1977](#); [Konrad et al., 2008](#)). Furthermore, a single woman may not have enough support or allies on the board to effectively represent the perspectives and priorities of women and other underrepresented groups ([Terjesen et al., 2009](#)). Therefore, having a critical mass of women on the board, rather than just one or a few, is likely to be more effective in ensuring that different perspectives and priorities are represented and considered in decision-making.

The critical mass theory suggests that having multiple women on the board, rather than just one or a few, is necessary for them to have a significant influence on decision-making ([Cambrea et al., 2023](#)). According to this theory, it is only when the number of women reaches a specific threshold, typically three or more, that they are able to counteract the

dominant culture and power dynamics of the board and make their voices heard on an equal footing to male board members (Joecks *et al.*, 2013; Torchia *et al.*, 2011). This proposition is confirmed by Post *et al.* (2011), de Masi *et al.* (2021) and Ben-Amar *et al.* (2017) who find that firms with three or more female directors have higher environmental and social performance as compared to other firms. Therefore, we posit the following proposition as follows:

- H2. Three or more women on the board are more likely to decrease a firm’s ESG controversies, whereas two or fewer women on the board are less likely to decrease a firm’s ESG controversies.

3. Methods

3.1 Sample

This study is based on a panel data set of non-financial enterprises operating in 13 European nations between 2004 and 2021. The initial sample consists of 10,891 firm-year observations. Because of the specific characteristics of financial firms’ disclosures and accounting processes, we omit them from the analysis. Firms with missing observations from 2004 to 2021 are also dropped. Countries with small observations are excluded from our sample. As a result, the final sample contains 9236 firm-year observations. Our data are provided by the Refinitiv Eikon database.

Tables 1 and 2 present the distribution of the sample and relevant variables (e.g. ESG controversies score and board gender diversity) by country and year. These tables likely provide a breakdown of the data by geographic location and period, allowing for analysis of trends and patterns in the data. Table 1 shows that the majority of the observations in the sample are from firms based in the UK (2,823), followed by Germany (1,085), France (1,078) and Switzerland (936). This suggests that the sample is heavily skewed toward firms from these countries, and may not be representative of firms from other countries. However, in the regression analysis, we excluded such countries to check the robustness of the results.

Table 1 indicates that the average value of ESG controversies score across the sample countries varied, with the highest average score of 93.082% recorded in Austria and the lowest average score of 84.467% recorded in Ireland. This suggests that firms based in

Country	N	Percentage	ESG controversies (%)	BGD (%)
Austria	195	2.11	93.082	18.867
Belgium	237	2.57	91.409	24.72
Denmark	344	3.72	92.556	21.089
Finland	459	4.97	90.91	27.544
France	1078	11.67	85.458	29.451
Germany	1,085	11.75	86.012	22.427
Ireland	373	4.04	84.467	17.712
Luxembourg	178	1.93	91.169	16.828
The Netherlands	491	5.32	89.472	19.801
Norway	246	2.66	88.887	39.128
Sweden	791	8.56	93.026	31.466
Switzerland	936	10.13	91.436	14.002
UK	2,823	30.57	88.041	18.871
All countries/mean	9,236	100	88.817	22.255

Table 1.
Sample distribution
by country

Source: Author’s own creation/work

				Reducing ESG controversies
Year	N	ESG controversies (%)	BGD (%)	
2004	190	87.158	7.104	
2005	296	88.548	8.578	
2006	319	89.61	10.417	
2007	322	88.7	10.199	
2008	346	86.077	9.950	
2009	304	86.473	11.434	
2010	386	84.732	13.369	
2011	418	84.048	15.266	
2012	426	85.856	17.201	
2013	431	86.334	19.253	
2014	453	86.065	21.226	
2015	537	91.784	23.224	
2016	574	91.548	25.335	
2017	596	88.877	27.070	
2018	780	90.974	28.087	
2019	900	90.907	29.26	
2020	990	89.006	29.939	
2021	968	90.929	31.331	
All years/mean	9,236	88.817	22.255	

Table 2.
Sample distribution by year (2004–2021)

Source: Author's own creation/work

Austria had the lowest number of controversies related to ESG issues among the sample countries, while firms based in Ireland had the highest number of controversies in relation to ESG issues. Column 5 in [Table 1](#) shows that on average, the highest representation of female directors on boards is found in Norway at 39.128%, and the lowest representation of female directors on boards is found in Switzerland at 14.002%.

Interestingly, [Table 2](#) shows a growing number of firm observations throughout the sample period (2004–2021). [Table 2](#) shows that the yearly average ESG controversies score in the sample firms ranges between 87.158% (2004) and 90.929% (2021), indicating that there is a significant increase in the reduction of controversies related to ESG issues. This suggests that over time, the firms in the sample have made efforts to reduce their negative impact on the environment and society, leading to an improvement in their ESG controversies score. It also implies a positive trend of firms becoming more sustainable and responsible over time. [Table 2](#) also shows an increase in the representation of female directors on boards over the sample period. The percentage of female board directors increases from 7.104% in 2004 to 31.331% in 2021. This suggests that over time, there has been a trend of increasing representation of women on boards in sample firms. This could be due to a greater emphasis on diversity and inclusion in the business world and a growing awareness of the benefits of having diverse perspectives in leadership positions.

3.2 Measures

3.2.1 Dependent variable. In this study, we use ESG metrics from the Refinitiv Eikon database. The ESG score for each individual company is generated by content research analysts at the Refinitiv Eikon database, using publicly available information sources. This implies that the score is not based on the self-reported data from the companies, but on the information that is publicly available and analyzed by the research analysts. This approach adds a level of objectivity and independence to the research ([Refinitiv Eikon Datastream, 2020](#)).

The ESG controversies score (ESGC_SCORE) is a metric used to measure a company's exposure to ESG controversies and negative events reflected in global media. It allows the researcher to understand the level of controversies a company has been associated with. A score of 100 indicates that a company has not been associated with any controversies, and a score lower than 100 indicates the presence of controversies. This metric provides insight into the reputation and sustainability of the company and can be used to compare companies within the same industry or sector ([Refinitiv Eikon Datastream, 2022](#)).

Using the Refinitiv Eikon database for ESG metrics offers several advantages for the research. The database's scoring methodology takes into consideration the country of incorporation and industry classifications of the companies, which allows for a more accurate and relevant assessment of their performance. This is important because companies operating in different countries and industries may have different regulatory and social expectations for their ESG performance. Additionally, the database uses a methodology that compares a company's ESG metrics to that of the same region and industry peers, which helps to maintain the objectivity of the assessment by considering the context and benchmarking it against similar companies. This approach allows for a more accurate comparison of company's performance and trends within the industry and region ([Refinitiv Eikon Datastream, 2020](#)).

3.2.2 Independent variables. The interest variable in this study is board gender diversity (BGD), which is defined as the percentage of women on the board of directors. The use of this measure is consistent with previous studies on board diversity, such as [Bear et al. \(2010\)](#), [Francoeur et al. \(2019\)](#) and [Issa and Bensalem \(2022\)](#). Using a consistent measure across studies allows for comparison of results and contributes to the body of knowledge on the topic. Additionally, measuring board diversity in terms of the percentage of women on the board is a widely used and recognized measure, which makes it easier to compare results with other studies and understand the meaning of the findings.

3.2.3 Control variables. We use a set of control variables that may have an impact on the firm's ESG controversies. First, a larger board size (B_SIZE) may permit the inclusion of a variety of perspectives on corporate strategy ([Rao and Tilt, 2021](#)), which can improve environmental and social performance. This is because a larger board size allows for a more diverse group of individuals with different backgrounds, experiences and perspectives to make decisions, which can lead to more effective and responsible decision-making. On the other hand, some researchers argue that a large number of board members may be ineffective due to free-rider issues and disagreements in the decision-making process ([Lipton and Lorsch, 1992](#)). Therefore, in this study, the board size is measured as the natural log of total number of board directors.

Second, the proportion of independent board members on a company's board of directors (B_IND). Independent directors are individuals who are not affiliated with the company or its management, and are expected to provide objective oversight and make decisions in the best interests of the company's stakeholders. Past studies suggest that board independence is associated with better ESG decisions ([Chang et al., 2017](#)). Thus, board independence (B_IND) is defined as the proportion of independent board members. Third, CEO duality (CEOD) is included in our model. When the CEO holds the position of chairman of the board, they have significant control and influence over the decision-making process of the board. Thus, CEO duality (CEOD) is a dummy variable in our model, and it takes 1 in the event that the CEO is also the chair, and 0 if this is not the case. Finally, board tenure (B_TENURE) is included as a control variable because the length of time that a director has served on the board can affect the board's decision-making process ([Harjoto et al., 2015](#)).

Firm-specific characteristics can also play an important role in a company's ESG performance. Therefore, we include a set of variables that are related to firm-specific characteristics. First, firm size (F_SIZE) can affect its environmental performance, as larger companies may have more resources available to invest in environmental initiatives and may be subject to different regulations than smaller companies (Elsayed, 2006). We control for firm size using the natural logarithm of total assets. Second, profitable firms may have more financial resources to devote to environmental projects and compliance with regulations, as well as greater access to capital for investments in new technologies or processes. Therefore, the return-of-asset ratio (ROA) is computed by dividing net income before extraordinary items by total assets. Third, firm's market valuation is often associated with its environmental performance (Kumar and Shetty, 2018). Therefore, market to book value ratio (M_TO_B) is used in our econometric model. Fourth, companies with larger leverage may be more sensitive to the demands of their creditors, who may be more likely to demand environmentally-friendly practices in order to reduce the risk of default. Leverage (LEV) is calculated as total debt divided by total equity. Fifth, industry dummies are used to control the different environmental challenges and regulations that companies in different industries may face. Finally, year dummies are used to control for unobservable factors such as changes in environmental regulations, economic conditions and societal expectations over time.

3.3 Econometric model

To examine the effect of board gender diversity (the independent variable) on a company's ESG controversies score (the dependent variable), we use several econometric models using the OLS) method:

$$ESGC_SCORE_{it} = \beta_0 + \beta_1 BGD_{it} + \sum_{i=0}^n \beta_n Control\ Variables_{it} + \varepsilon_{it} \quad (1)$$

$$ESGC_SCORE_{it} = \beta_0 + \beta_1 FEM3_{it} + \sum_{i=0}^n \beta_n Control\ Variables_{it} + \varepsilon_{it} \quad (2)$$

$$ESGC_SCORE_{it} = \beta_0 + \beta_1 FEM2_{it} + \sum_{i=0}^n \beta_n Control\ Variables_{it} + \varepsilon_{it} \quad (3)$$

where ESGC controversies score (ESGC_SCORE) in three equations represents our dependent variable. BGD is a measure of the percentage of female directors on a company's board of directors. FEM3 and FEM2 are binary variables that are used to indicate the presence of a certain number of female directors on the board (e.g. three or more and two or one female directors on the board, respectively). Control refers to a set of variables that could potentially influence the ESG controversies by the sample firms. Table 3 provides the measurements of all the variables used.

4. Results

Table 4 reports descriptive statistics for the sample. The firm's ESG controversies score (ESGC_SCORE) is a measure of a firm's performance with regard to ESG controversies, and the average score for the sample is 88.817%. This means that, on average, the firms in the sample have a relatively high level of performance with regard to ESG controversies. On average, gender diversity is 22.256% on corporate boards, whereas the maximum is 80%. This result is slightly lower than the findings of Nuber and Velte (2021) who reported a

Variable	Symbol	Definition
<i>Dependent variable</i>		
ESG controversies score	ESGC_SCORE	It measures a company's exposure to ESG controversies and negative events reflected in global media. A score of 100 indicates that a company has not been associated with any controversies, and a score lower than 100 indicates the presence of controversies
<i>Independent variable</i>		
Board gender diversity	BGD	The ratio of female directors on the board compared with the total number of board directors
Three or more females on the board	FEM3	A dummy variable capturing the presence of three or more females on the board
Two or less females on the board	FEM2	A dummy variable capturing the presence of two or one female on the board
<i>Control variables</i>		
Board size	B_SIZE	The natural logarithm of total number of board members
Board independence	B_IND	The proportion of independent directors to the total number of board members
CEO duality	CEOD	A dummy variable coded 1 if the CEO simultaneously chair the board been the CEO of the company, and 0 otherwise
Board tenure	B_TENURE	The length of board members' tenure
Firm size	F_SIZE	The natural logarithm of the firm's total assets
Profitability	ROA	The ratio of net income divided by total assets
Market to book value ratio	M_TO_B	The ratio of a firm's book value to its market value
Leverage	LEV	The ratio of total debt divided by total equity
Industry dummy	INDUSTRY	A dummy variable represents 10 industries
Year dummy	YEAR	A dummy variable represents the period of 2004–2021

Source: Author's own creation/work

Table 3.
Variables and definitions

Variable	Obs	Mean	SD	Min	Max
ESGC_SCORE	9,236	88.817	24.84	0.806	100
BGD	9,236	22.256	14.844	0	80
FEM3	9,236	0.288	0.453	0	1
FEM2	9,236	0.712	0.453	0	1
B_SIZE	9,236	9.946	3.602	1	28
B_IND	9,236	58.742	25.819	0	100
CEOD	9,236	0.226	0.419	0	1
B_TENURE	9,236	6.247	2.758	0.083	31.313
F_SIZE	9,236	22.358	1.577	15.98	27.137
ROA	9,236	0.065	0.107	−2.072	2.493
M_TO_B	9,236	1.951	4.708	0.002	109.554
LEV	9,236	1.711	15.594	−830.25	616.667

Table 4.
Descriptive statistics

Source: Author's own creation/work

mean value of board gender diversity 23.27% in a sample of the European STOXX600 index. Table 5 reports the results of correlation analysis and variance inflation factors (VIFs). The results of the correlation analysis indicate that there is no significant correlation between any of the variables in the data set, suggesting there are no issues with multicollinearity. Furthermore, the VIF values are less than 5, indicating no multicollinearity issues in the data set (O'Brien, 2007).

Table 6 reports the results of a statistical analysis testing the link between board gender diversity and ESG controversies score. Model 1 shows that the coefficient of BGD is positive and significant ($\beta = 0.055, p < 0.01$), suggesting that the presence of female members on the board is positively associated with ESG controversies score. This means that as the proportion of female members on the board increases, the ESG controversies score also increases. Therefore, this result supports *H1*, which states that the presence of female members on the board is positively associated with the ESG controversies score.

We run additional tests to check whether any changes in the relationship between the variables are a result of changes in the independent variable (board gender diversity) or changes in other factors. Model 2 reports results with excluding control variables related to board characteristics. The board of directors is responsible for making major decisions within a company, and it is likely that the board plays an active role in improving the ESG controversies score. Therefore, we exclude board-level variables in Model 2 to eliminate the board characteristics impact. The findings in Model 2 indicate that the presence of female members on the board may have a direct influence on the organization's performance in terms of ESG controversies. Moreover, the process of change in a firm can be incremental, and it may take time for female directors to gain traction on the board and to change corporate policies. Therefore, in Model 3, we use a one-year lagged board gender diversity variable to determine if our findings remain consistent over time. The results in Model 3 suggest that the relationship between board gender diversity and ESG controversies score is stable over time. Furthermore, to determine if the findings are influenced by a few countries that have a large representation in the sample, we re-estimate the analysis after excluding the UK, Germany, France and Switzerland firms in Model 4. Results in Model 4 suggest the findings are robust, and that the relationship between the variables is likely generalizable to other countries.

In Models 5 and 6, we test the link between the critical mass of female directors, as measured by the binary variables (e.g. three or more female directors (FEM2) and two or one female directors (FEM1)), and the ESG controversies score. This analysis can help to understand if the presence of a critical mass of female directors is important to achieve a higher ESG controversies score, and how the number of female directors affects the performance of the company in terms of ESG controversies. Model 5 shows that the coefficient of FEM2 is positive and significant ($\beta = 2.619, p < 0.01$), whereas Model 6 indicates FEM1 is insignificant. These results indicate that the presence of three or more female directors on the board is positively associated with a higher ESG controversies score, whereas the presence of two or one female director on the board has no statistically significant relationship with the ESG controversies score. These results support our *H2* and *H3*. The results support the idea that a certain threshold of women's participation on boards is necessary for them to have an equal voice and influence as men. The results support the critical mass theory, suggesting a minimum percentage of women on boards is needed for them to have a meaningful impact on the decision-making process and to avoid the phenomenon of tokenism.

4.1 Robustness checks

We perform several tests to ensure the stability of results. First, we use alternative measurement of ESG controversies. A dummy variable is constructed indicating 1 if the

Table 5.
Pairwise correlations

Variables	VIF	1/VIF	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) ESGC_SCORE			1.000											
(2) BGD	2.71	0.369	0.043***	1.000										
(3) FEM3	2.69	0.371	0.053***	0.796***	1.000									
(4) FEM2	2.69	0.371	0.013	-0.796***	-1.000	1.000								
(5) B_SIZE	1.46	0.685	-0.180***	-0.062***	-0.108***	0.108***	1.000							
(6) B_IND	1.14	0.875	-0.052***	0.082***	0.053***	-0.053***	-0.178***	1.000						
(7) CEOD	1.03	0.967	-0.034	-0.048***	-0.040***	0.040***	0.036*	-0.036*	1.000					
(8) B_TENURE	1.08	0.928	-0.024	-0.102***	-0.099***	0.099***	-0.102***	-0.102***	0.156***	1.000				
(9) F_SIZE	1.38	0.723	-0.325***	-0.060***	-0.084***	0.084***	0.072***	0.052***	0.130***	0.130***	1.000			
(10) ROA	1.06	0.941	0.036***	-0.020	-0.051***	0.051***	-0.019	0.057***	0.071***	0.071***	0.111***	1.000		
(11) M_TO_B	1.12	0.892	0.037***	-0.090***	-0.063***	0.063***	-0.054***	-0.038**	0.013	0.020	-0.044***	-0.047***	1.000	
(12) LEV	1.09	0.897	-0.015	0.024*	0.026*	-0.026	0.006	0.029	-0.017	-0.016	-0.005	-0.003	-0.001	1.000

Notes: This table presents the correlation analysis. All variables are defined in Table 3, *significant at 10% level, **significant at 5% level, ***significant at 1% level

Source: Author's own creation/work

Table 6.
Board gender
diversity and critical
mass vs ESG
controversies score
using OLS
regressions

Variable	Model 1	Model 2 Excluding board-level variables	Model 3 Lag BGD	Model 4 Excluding the UK, Germany France and Switzerland firms	Model 5 Critical mass	Model 6 Critical mass
BGD	0.055*** (3.18)	0.037** (2.27)	0.058*** (3.05)	0.063*** (3.64)	2.619*** (4.79)	0.619 (1.79)
Fem3						-1.755** (-2.45)
FEM2						-0.015* (-1.69)
B_SIZE	-1.92*** (-2.63)		-2.042** (-2.55)	-0.19 (0.17)	-1.755** (-2.45)	-0.015* (-1.69)
B_IND	-0.016* (-1.78)		-0.016 (-1.62)	-0.02 (-1.24)	-0.015* (-1.69)	1.127* (1.89)
CEO	1.187** (1.99)		1.009 (0.116)	-0.567 (-0.55)	1.127* (1.89)	0.173** (2.20)
B_TENURE	0.174** (2.21)		0.298*** (3.37)	0.051 (0.36)	0.173** (2.20)	-6.806*** (-30.90)
F_SIZE	-6.88*** (-30.79)	-6.969*** (-36.37)	-7.201*** (-29.72)	-5.854*** (-15.89)	-6.806*** (-30.90)	-1.266 (-0.65)
ROA	-1.353 (-0.70)	-1.442 (-0.76)	-1.471 (-0.65)	7.65** (2.35)	-1.266 (-0.65)	-0.167*** (-3.20)
M_TO_B	-0.164*** (-3.14)	-0.155*** (-3.43)	-0.192*** (-2.94)	-0.135 (-1.62)	-0.167*** (-3.20)	-0.009 (-0.48)
LEV	-0.009 (-0.48)	-0.008 (-0.41)	-0.009 (-0.48)	-0.010 (-0.36)	-0.009 (-0.48)	232.809*** (25.08)
Constant	230.479*** (24.85)	232.219*** (25.51)	243.903*** (24.78)	202.783*** (17.12)	230.19*** (24.98)	Yes
Industry	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Observation	9,236	9,831	8,309	3,317	9,236	9,236
R-squared	0.217	0.211	0.221	0.214	0.218	0.208
F (p-value)	48.060 (0.000)	55.385 (0.000)	47.375 (0.000)	20.592 (0.000)	48.098 (0.000)	48.091 (0.000)

Notes: This table presents the OLS. Reported results include *t*-test in parentheses along with coefficients. All standard errors are robust to control for heteroscedasticity. All variables are defined in Table 3. *significant at 10% level, **significant at 5% level, ***significant at 1% level

Source: Author's own creation/work

firm has ESG controversies, and 0 otherwise. We use probit regression in Table 7, which is a commonly used technique for modeling dichotomous or binary dependent variables. The results in Table 7 are similar to the findings of our main model. Second, some researchers suggest that high-polluting firms may be more likely to take proactive measures to address their environmental concerns, such as implementing sustainable practices, reducing emissions and increasing transparency, to mitigate the risk of negative consequences from stakeholders (Liao *et al.*, 2015). Therefore, we class the sample into non-GHG-intensive firms and GHG-intensive firms in Table 8 and rerun our baseline model. Following Lund (2007), firms that belong to the basic materials, energy, industrials, and utilities sectors are classified as GHG-intensive firms, as they are more likely to have higher GHG emissions due to their operations. On the other hand, firms that belong to the remaining consumer discretionary, consumer staples, health care, real estate, technology and telecommunications sectors are classified as non-GHG-intensive firms, as they are less likely to have high emissions. Interestingly, results in Models 3 and 4 show that the presence of women on the board is positively and significantly associated with better ESG performance in both subsamples of non-GHG-intensive and GHG-intensive firms. Third, firms with strong governance structures may also have a greater focus on transparency and accountability, which could make it easier to measure and report on ESG controversies. Therefore, we control for a potential confounding factor such as firm-level governance quality by comparing the results of firms with weak governance performance to those with strong governance performance. In Table 9, we split the sample into weak and strong governance performance based on Eikon corporate governance score. Fourth, as some firms may be more environmentally conscious than others, we investigate whether our findings depend on the degree of environmental orientation of the firms. In Table 10, we split the sample into low and high environmental performance based on Eikon corporate environmental score.

Variable	Model 1	Model 2 Critical mass	Model 3 Critical mass
BGD	−0.002** (−2.04)		
Fem3		−0.142*** (−3.76)	
FEM2			0.032 (0.76)
B_SIZE	0.143** (2.34)	0.136** (2.23)	0.136** (2.23)
B_IND	0.002** (2.18)	0.002** (2.22)	0.002** (2.22)
CEOD	−0.086** (−2.03)	−0.081* (−1.92)	−0.081* (−1.92)
B_TENURE	−0.003 (−0.44)	−0.003 (−0.46)	−0.003 (−0.46)
F_SIZE	0.535*** (29.70)	0.535*** (29.63)	0.535*** (29.63)
ROA	0.209 (0.70)	0.193 (0.65)	0.193 (0.65)
M_TO_B	0.011** (1.96)	0.011* (1.92)	0.011* (1.92)
LEV	0 (0.08)	0 (0.08)	0 (0.08)
Constant	−13.198*** (−26.02)	−13.178*** (−26.04)	−13.32*** (−26.15)
Industry	Yes	Yes	Yes
Year	No	No	No
Observation	9,236	9,236	9,236
Pseudo R	0.2791	0.2801	0.280
Chi2 (p-value)	1,536.57 (0.000)	1,539.34 (0.000)	1,539.344 (0.000)

Table 7.
ESG controversies
dummy probit
regressions

Notes: This table presents the probit results. Reported results include *t*-test in parentheses along with coefficients. All standard errors are robust to control for heteroscedasticity. All variables are defined in Table 3; *significant at 10% level; **significant at 5% level; ***significant at 1% level

Source: Author's own creation/work

Variable	Non-GHG-intensive firms		GHG-intensive firms	
	Model 1	Model 2	Model 3	Model 4
BGD	0.064*** (2.96)			0.044** (1.83)
Fem3		2.895*** (4.18)		
FEM2			-0.095 (-0.18)	
B_SIZE	-2.598*** (-2.75)	-2.418*** (-2.61)	-2.418*** (-2.61)	-0.856 (-0.73)
B_IND	-0.029** (-2.46)	-0.028** (-2.38)	-0.028** (-2.38)	-0.011 (-0.80)
CEO	0.791 (0.96)	0.802 (0.98)	0.802 (0.98)	1.513* (1.73)
B_TENURE	0.044 (0.42)	0.032 (0.31)	0.032 (0.31)	0.323*** (2.63)
F_SIZE	-7.088*** (-24.89)	-7.096*** (-24.97)	-7.096*** (-24.97)	-6.418*** (-18.06)
ROA	-1.114 (-0.54)	-1.038 (-0.50)	-1.038 (-0.50)	-0.507 (-0.10)
M_TO_B	-0.221*** (-3.84)	-0.225*** (-3.91)	-0.225*** (-3.91)	0.174 (1.32)
LEV	0.002 (0.08)	0.002 (0.08)	0.002 (0.08)	-0.033 (-1.31)
Constant	242.253*** (23.36)	242.074*** (23.48)	244.97*** (23.55)	217.261*** (19.58)
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Observation	4,996	4,996	4,996	4,259
R-squared	0.255	0.257	0.159	0.184
F (p-value)	34.946 (0.000)	35.067 (0.000)	29.013 (0.000)	21.608 (0.000)
				217.018*** (19.79)
				219.475*** (19.74)
				Yes
				Yes
				4.259
				0.139
				19.261 (0.000)

Notes: This table presents the OLS. Reported results include *t*-test in parentheses along with coefficients. All standard errors are robust to control for heteroscedasticity. All variables are defined in Table 3; *significant at 10% level; **significant at 5% level; ***significant at 1% level

Source: Author's own creation/work

Table 8.
Controlling for firm-level industry using OLS regressions

Table 9.
Controlling for firm-
level corporate
governance using
OLS regressions

Variable	High corporate governance performance			Low corporate governance performance		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	Critical mass	Critical mass	Critical mass	Critical mass	Critical mass	Critical mass
BGD	0.153*** (5.33)	2.029*** (2.97)	-0.029 (-0.57)	0.052* (1.32)	2.649** (3.61)	-0.049 (-0.71)
FEM3			-2.729** (-2.44)	-1.052 (-1.20)	-0.533 (-0.63)	-0.533 (-0.63)
FEM2	-2.829** (-2.52)	-0.009 (-0.64)	-0.009 (-0.64)	-0.011 (-1.05)	-0.007 (-0.65)	-0.007 (-0.65)
B_SIZE	-0.009 (-0.65)	1.347* (1.75)	1.347* (1.75)	-0.508 (-0.59)	-0.613 (-0.71)	-0.613 (-0.71)
B_IND	1.357* (1.76)	0.161 (1.39)	0.161 (1.39)	0.268*** (2.73)	0.263*** (2.68)	0.263*** (2.68)
CEO_D	0.162 (1.40)	-6.804*** (-24.62)	-6.804*** (-24.62)	-5.915*** (-16.58)	-5.95*** (-16.72)	-5.95*** (-16.72)
B_TENURE	-6.815*** (-24.58)	-1.357 (-0.63)	-1.357 (-0.63)	-2.335 (-0.62)	-2.511 (-0.66)	-2.511 (-0.66)
F_SIZE	-1.512 (-0.70)	-0.176*** (-2.98)	-0.176*** (-2.98)	-0.147 (-1.59)	-0.148 (-1.58)	-0.148 (-1.58)
ROA	-0.172*** (-2.92)	0 (0.01)	0 (0.01)	-0.019 (-0.72)	-0.02 (-0.77)	-0.02 (-0.77)
M_TO_B	0 (0.00)	227.496*** (21.81)	229.525*** (21.85)	229.651*** (12.79)	229.831*** (12.88)	234.48*** (13.04)
LEV	228.174*** (21.70)	Yes	Yes	Yes	Yes	Yes
Constant	Yes	Yes	Yes	Yes	Yes	Yes
Industry	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Observation	6,391	6,391	6,391	3,280	3,280	3,280
R-squared	0.221	0.221	0.181	0.201	0.204	0.190
F (p-value)	36.923 (0.000)	36.917 (0.000)	30.110 (0.000)	12.040 (0.000)	12.138 (0.000)	10.128 (0.000)

Notes: This table presents the OLS. Reported results include *t*-test in parentheses along with coefficients. All standard errors are robust to control for heteroscedasticity. All variables are defined in Table 3; *significant at 10% level; **significant at 5% level; ***significant at 1% level

Source: Author's own creation/work

The results of the subsample analysis, as presented in [Tables 9 and 10](#), indicate that the coefficient of board gender diversity is stronger in companies with strong governance systems and better environmental performance. This suggests that the link between board gender diversity and ESG controversies score may be more pronounced in firms that are already performing well in terms of governance and environmental performance. This could indicate that firms with strong governance and environmental performance may be better equipped to take advantage of the potential benefits of diversity in decision-making.

Fifth, our results provide evidence that female board directors have an impact on the ESG controversies score. However, it is also possible that the relationship between female directors and ESG controversies score may be influenced by ESG performance of the firm. Specifically, it is possible that firms that have adopted high levels of ESG initiatives are the ones that have a higher representation of women on their boards. This suggests that firms that are already committed to environmental performance may be more likely to prioritize diversity in decision-making, including hiring more female directors. If this is the case, our results may suffer from reverse causality or simultaneity issue. To address this concern, we use the two-stage least squares (2SLS) method with the industry level of female board members as external instrumental variables. The industry level of female board members is a variable that is not directly affected by the firm's ESG controversies score, but it may impact the corporate board gender diversity (endogenous variable) at the firm level ([Issa and Zaid, 2021](#); [Nadeem et al., 2020](#); [Zalata et al., 2019](#)). As shown in [Table 11](#), the estimates obtained using the 2SLS method with industry level of female board members as instrumental variables are similar to the estimates obtained using the basic model without controlling for endogeneity. This concordance confirms that the reverse causality/simultaneity issue does not significantly affect the results of your analysis.

5. Discussion

Modern companies are under significant pressure to address environmental and social concerns due to increased awareness among shareholders and pressure from stakeholders. ESG controversies can have a negative impact on a company's valuation and performance. For instance, if a company is involved in an environmental scandal, it may face financial penalties and damage to its reputation, which can lead to a decline in stock value and loss of customers. Additionally, if a company is perceived as not addressing social or governance issues, it may face pressure from investors and stakeholders, which can also negatively affect its performance. Therefore, companies that prioritize and effectively manage ESG risks are likely to have better long-term financial performance than those that do not.

Despite the increasing importance of ESG controversies for modern companies, there is a scarcity of empirical studies examining the relationship between ESG controversies and board gender diversity. This study aims to address the lack of research in this area by investigating the relationship between board gender diversity and ESG controversies using a panel data set of non-financial enterprises operating in 13 European nations between 2004 and 2021. The study finds that having more female directors on the board leads to fewer ESG controversies for the firm. This effect is most pronounced when there are three or more women in the boardroom, which supports the critical mass theory. Moreover, this relationship may be influenced by several factors such as industry, effective governance system and environmental performance. The results are consistent with the use of different econometric specifications, alternative measures of ESG controversies and subsample analysis.

The results are in alignment with assumptions of the gender socialization theory, diversity theory and risk aversion literature. According to the gender socialization theory, women may be socialized to be more sensitive to ethical issues such as ESG, which may lead

Table 10.
Controlling for firm-
level environmental
performance using
OLS regressions

Variable	High environmental performance			Low environmental performance		
	Model 1	Model 3	Model 4	Model 5	Model 5	Model 6
BGD	0.076*** (5.15)	1.131*** (2.94)		0.024* (2.77)	2.04** (1.26)	
FEM3			0.621* (1.04)			0.041 (0.126)
FEM2		1.855*** (2.92)	1.855*** (2.92)	1.554** (2.17)	2.282*** (3.20)	2.282*** (3.20)
B_SIZE	1.545** (2.43)	0.019*** (2.60)	0.019*** (2.60)	0.007 (0.75)	0.015* (1.72)	0.015* (1.72)
B_IND	0.016** (2.16)	1.086*** (2.68)	1.086*** (2.68)	0.299 (0.54)	0.308 (0.56)	0.308 (0.56)
CEOD	0.997** (2.47)					
B_TENURE	-0.075 (-1.11)	-0.104 (-1.54)	-0.104 (-1.54)	-0.061 (-0.80)	-0.096 (-1.27)	-0.096 (-1.27)
F_SIZE	3.792*** (25.53)	3.788*** (25.47)	3.788*** (25.47)	3.712*** (18.41)	3.749*** (18.52)	3.749*** (18.52)
ROA	6.429** (2.28)	6.928** (2.51)	6.928** (2.51)	8.702*** (5.73)	9.38*** (6.20)	9.38*** (6.20)
M_TO_B	0.161* (1.74)	0.169* (1.86)	0.169* (1.86)	-0.006 (-0.15)	-0.01 (-0.26)	-0.01 (-0.26)
LEV	-0.002 (-0.42)	-0.002 (-0.55)	-0.002 (-0.55)	0.006 (0.19)	0.009 (0.30)	0.009 (0.30)
Constant	-45.257*** (-10.16)	-46.079*** (-10.32)	-44.949*** (-10.05)	-59.489*** (-11.24)	-62.522*** (-11.85)	-58.482*** (-10.87)
Industry	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
Observation	4,878	4,878	4,878	4,358	4,358	4,358
R-squared	0.233	0.230	0.201	0.243	0.214	0.193
F (p-value)	49.867 (0.000)	49.395 (0.000)	40.201 (0.000)	46.318 (0.000)	42.057 (0.000)	39.127 (0.000)

Notes: This table presents the OLS. Reported results include *t*-test in parentheses along with coefficients. All standard errors are robust to control for heteroscedasticity. All variables are defined in Table 3. *significant at 10% level; **significant at 5% level; ***significant at 1% level

Source: Author's own creation/work

Variables	ESG controversies score		ESG controversies dummy	
	Model 1	Model 2	Model 3	Model 4
		Critical mass	Critical mass	Critical mass
BGD	0.079*** (3.10)	3.76*** (4.58)		-0.054*** (-3.98)
Fem3				
FEM2			0.76 (0.58)	
B_SIZE	-2.158*** (-2.68)	-1.969** (-2.49)	-1.969** (-2.49)	0.045*** (3.34)
B_IND	-0.017* (-1.75)	-0.017* (-1.71)	-0.017* (-1.71)	-0.055*** (3.10)
CEOD	0.989 (1.54)	0.891 (1.39)	0.891 (1.39)	-0.015 (2.58)
B_TENURE				-0.002 (-1.41)
F_SIZE	-7.206*** (-29.76)	-7.22*** (-29.88)	-7.22*** (-29.88)	0.134*** (39.85)
ROA	-1.574 (-0.70)	-1.463 (-0.64)	-1.463 (-0.64)	0.089** (2.34)
M_TO_B	-0.191*** (-2.93)	-0.199*** (-3.04)	-0.199*** (-3.04)	0.003*** (3.55)
LEV	-0.009 (-0.47)	-0.009 (-0.47)	-0.009 (-0.47)	0.011 (-0.62)
Constant	244.203*** (24.76)	244.04*** (24.96)	247.8*** (25.04)	-2.772*** (-23.11)
Industry	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Observation	8,298	8,298	8,298	8,298
R-squared	0.221	0.222	0.222	0.276
Chi2 (p-value)	1,666.450 (0.000)	1,658.130 (0.000)	1,668.175 (0.000)	3,454.585 (0.000)
				3,454.585 (0.000)

Notes: This table presents the second-stage regression of 2SLS estimations. Reported results include *t*-test in parentheses along with coefficients. All variables are defined in [Table 3](#); *significant at 10% level; **significant at 5% level; ***significant at 1% level

Source: Author's own creation/work

Table 11.
2SLS regression

Reducing ESG
controversies

them to prioritize ESG concerns and advocate for more responsible corporate behavior (Ben-Amar *et al.*, 2017; Katmon *et al.*, 2019). This, in turn, may lead to fewer ESG controversies for the firm. Furthermore, diversity theory suggests that having a diverse group of individuals on the board can lead to better decision-making and problem-solving, as different perspectives are brought to the table (Mateos de Cabo *et al.*, 2012; Siciliano, 1996). This diversity of thought may also contribute to a more thorough consideration of ESG risks and opportunities, leading to fewer controversies. Moreover, risk aversion literature suggests that women may be more risk-averse than men. Women have been traditionally socialized to be more cautious and protective, which may lead them to avoid risk (Boulouta, 2013; Faccio *et al.*, 2016). Moreover, women tend to have different cognitive styles, which lead them to be more detail-oriented and to focus on the potential negative consequences of a decision, which may make them more risk-averse (Arnaboldi *et al.*, 2021; Dadanlar and Abebe, 2020; Hutchinson *et al.*, 2015). Therefore, the presence of women on boards can lead them to prioritize risk management and risk reduction. This may also translate to a more proactive approach to addressing ESG risks, leading to fewer controversies for the firm.

6. Conclusion

This study investigates the impact of board gender diversity on ESG controversies using data from non-financial European companies between 2004 and 2021. It finds that more women on the board lead to fewer ESG controversies, especially when there are three or more female directors. The relationship may also be influenced by factors such as industry, governance and environmental performance.

This study provides important implications for companies, policymakers, investors and stakeholders. For businesses, it suggests that increasing the representation of women on boards may help to mitigate ESG controversies and improve the overall reputation and performance of the firm. This may be particularly relevant for firms operating in industries where ESG risks are prevalent, such as those related to the environment, human rights and social issues. For policymakers, the findings may support the case for implementing policies and initiatives aimed at increasing the representation of women on boards. This may include setting targets for women's representation on boards, offering training and mentoring programs for women in leadership roles and encouraging firms to adopt more inclusive recruitment and promotion practices. Additionally, the results may also help investors, consumers and other stakeholders to better understand the relationship between board gender diversity and ESG controversies, which may be particularly important in light of the growing focus on ESG issues.

We acknowledge that this study has several limitations. First, this study uses a sample of firms operating in European nations, which means that the findings may not be generalizable to other regions. To expand on this research, future studies could examine firms in other regions, particularly in developing countries, to determine if the impact of having women on boards is consistent across different contexts. Second, this study does not incorporate moderating variables. Future studies could test some moderating variables such as the presence of a sustainability/CSR committee in a firm.

Note

1. The CDP is an international non-profit organization with offices in several countries, including the UK, Japan, India, China, Germany and the USA. The organization helps investors, companies, cities and states to analyze and manage their environmental impact by providing a standardized reporting platform for disclosure of climate-related information.

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